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Evaluation of water depth and inundation duration on *Typha domingensis* sustainability in the Everglades Stormwater treatment areas: A test cell study

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ABSTRACT

Typha domingensis is a dominant plant species in the Everglades Stormwater Treatment Areas, freshwater wetlands constructed to reduce phosphorus (P) levels in stormwater runoff prior to discharge into the Everglades. Water depth and duration of inundation can negatively affect the health of these plants. To determine the effect of water depth and inundation duration, *Typha* plants were grown in this study in a series of 15 test cells, 0.2 ha in area, and exposed to five water-depth treatments over 10 months: control (40 cm), shallow (61 cm), moderate (84 cm), deep (104 cm) and extremely deep (124 cm). Mean adult *Typha* densities declined from 14.1 shoots m^{-2} in the control to 10.9 and 10.5 shoots m^{-2} in the deep and extremely-deep treatments, respectively. However, only mean juvenile *Typha* densities were significantly different among water-level treatments, declining from 3.2 juveniles m^{-2} in the control to 1.2 and 0.6 juveniles m^{-2} in the deep and extremely-deep treatments, respectively. *Typha* leaf elongation rate (LER) was significantly higher in the deeper treatments, with the highest rates measured during the active growing season (July to October). LER increased from 6.84 cm day^{-1} during June baseline measurements to 8.89 cm day^{-1} in the deeper treatments after eight weeks of continuous inundation. *Typha* showed morphological plasticity to adapt to deep water through changes in morphology and biomass allocation. *Typha* grew significantly taller and heavier to adapt to deep-water conditions and facilitate gas exchange between the above and belowground tissues. Ramet height of *Typha* harvested at the end of the study averaged 343 and 374 cm in the deep and extremely-deep treatments, respectively, compared to 285 cm in the control. Likewise, the weight of individual adult ramets was significantly greater in the deeper treatments (256 g $ramet^{-1}$) compared to the control (123 g $ramet^{-1}$). Although none of the water-level treatments resulted in a total collapse of their *Typha* populations, increased LER in the deeper treatments resulted in less structural support in stems, causing substantial *Typha* lodging when water levels were lowered at the end of the study. These findings provide valuable insight into the effect of different water-level inundations on plant responses and overall health of *Typha* populations in the STAs.

1. Introduction

Emergent aquatic vegetation (EAV) is an essential component of treatment wetlands because they sequester nutrients from the water column in their tissues which eventually become part to the wetland soil after plant senescence (Vymazal, 2007; Chen, 2011; Bansal et al., 2019). Dense EAV growth acts as a physical barrier reducing flow velocities and facilitating settling of particles that may improve water quality (Chimney and Newman, 2006). EAV also provides structure for the attachment of periphyton that directly removes nutrients from the water column. *Typha domingensis* (hereafter simply *Typha*) is the most common emergent macrophyte in the Everglades Stormwater Treatment Areas (STAs) managed by the South Florida Water Management District (SFWMD). The STAs are constructed freshwater wetlands that reduce phosphorus (P) levels in stormwater runoff prior to this water being discharged to the Everglades (Chimney and Goforth, 2001; Sklar et al.,

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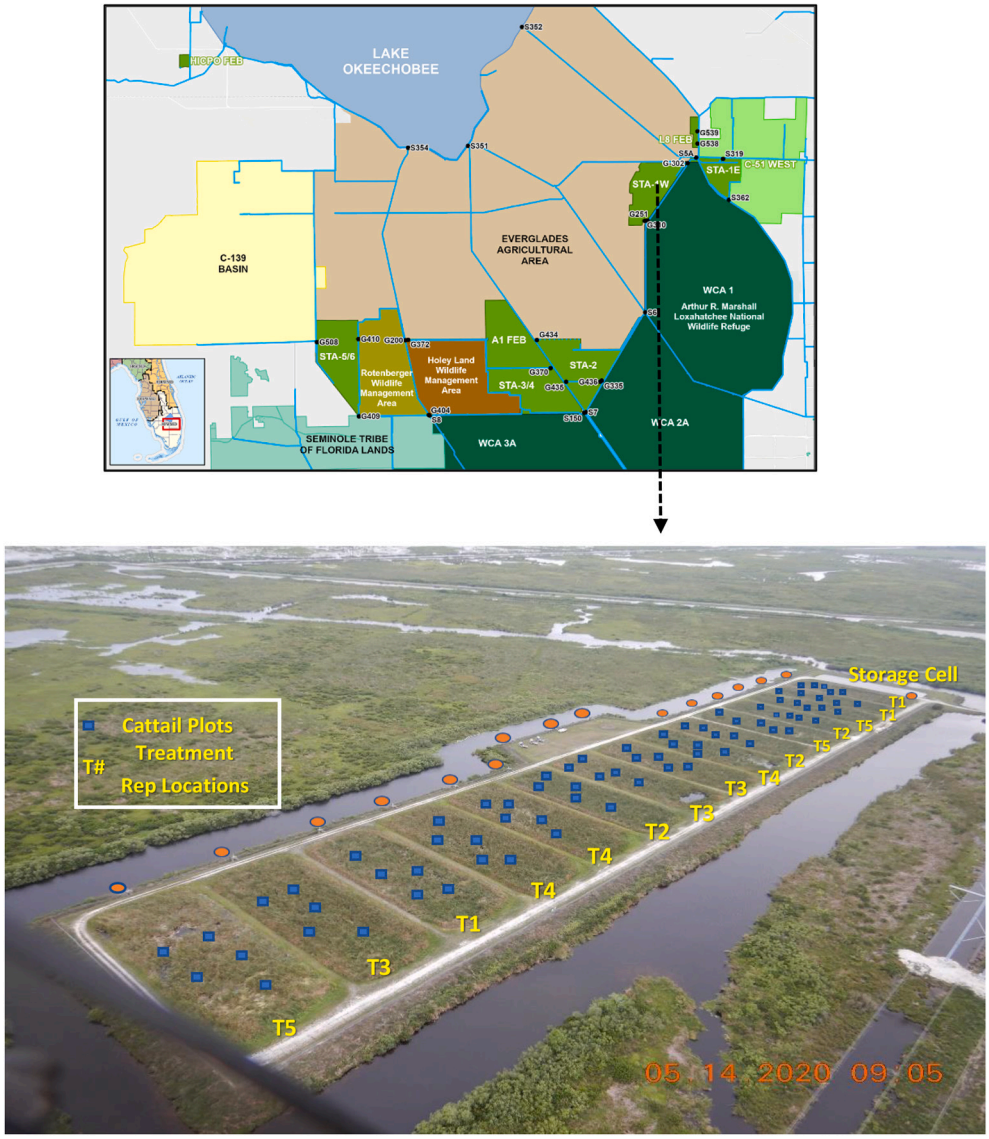


Fig. 1. Maps showing the location of STA-1 W test cells in South Florida and a photo of the test cells with approximate location of Typha plots and location of treatment replications.

2005; Armstrong et al., 2023). To provide effective nutrient removal in the STAs, it is important to understand and manage factors that affect vegetation health.

Typha spp. are obligate wetland plants that exist over a wide range of hydrological conditions (Newman et al., 1998; Cronk and Fennessy, 2001). *Typha* spp. can grow under flooded conditions because of their extensive aerenchyma or intercellular gas spaces, which transport oxygen from the shoots to below ground tissues (Armstrong, 1979; Armstrong et al., 1994). The aerenchyma produces a pressurized flow of gases that maintains aerobic conditions within the root zone, allowing the rhizomes and roots to grow (Brix et al., 1992; Bendix et al., 1994; Chabbi et al., 2000). *Typha* spp. also have the morphological plasticity to adapt to deep water. These adjustments include the increase of biomass allocation to aboveground tissues (Grace, 1989; Chen et al., 2010) and ability to increase plant height through rapid elongation of shoots, which facilitates gas exchange between the plant and the atmosphere (Blom and Voesenek, 1996; Grace, 1989). As water depths increase, *Typha* spp. grow taller, producing fewer but larger ramets (Grace, 1989). However, rapid shoot elongation and increase in aboveground biomass due to rising water levels requires sufficient belowground biomass for adequate anchorage and nutrient uptake (Grace, 1989; Miao and Zou, 2012).

Inundation exceeding optimal depth and duration increases physiological stress on *Typha,* reduces its effectiveness to store nutrients, and can result in plant mortality (Grace and Wetzel, 1982). Studies reported optimal growth of *Typha* at shallow water depths (~22-cm), producing significantly greater biomass than plants grown in deeper water (Grace, 1989; Waters and Shay, 1992; Miao and Zou, 2012). Others suggest that the optimal water depth for *Typha* ranges from 30 to 60 cm (Newman et al., 1998; Grace and Wetzel, 1998; White et al., 2007; Miao, 2014). Grace (1989) and Miao and Zou (2012) reported *Typha* being stressed at water depths >60 cm when inundated from a few weeks to more than a year, with seedlings and juveniles more affected by deep water than adult plants (Diaz and Vaughan, 2019). Increasing inundation depths from 40 cm to 91 and 137 cm for six weeks in a mesocosm study resulted in significant decreases of shoot density, biomass, and photosynthesis, altering the allocation of photosynthetic products, which is an important survival strategy (Pezeshki, 2001; Chen et al., 2010). Following a recovery period of four weeks, most adverse effects that occurred at 91 cm water depth were reversed, while the effects of the 137 cm water depth were lethal to all *Typha* plants. Increasing water depth also reduces the anchorage capacity of *Typha* as plants respond to deeper water by

decreasing biomass allocation to roots and rhizomes and increasing allocation to shoots (Grace, 1989; Chen et al., 2010). Prolonged and repeated periods of deep-water in the STAs have resulted in decrease of areal coverage and poor *Typha* growth leading to the formation of floating tussocks (Chimney et al., 2000; Chen et al., 2010; Diaz and Vaughan, 2019).

Fluctuating water levels are common in wetlands and depending on the depth amplitude, can influence the growth and productivity of EAV (Harris and Marshall, 1963; Deegan et al., 2007). Information on how EAV growth is related to changes in water depth is vital to understanding and predicting the responses of macrophytes such as *Typha* spp. to water-level manipulations (Froend and McComb, 1994; Chen et al., 2010) and important for managing *Typha* populations in the STAs. The objective of this study was to investigate the effect of different water-level treatments and duration of inundation on *Typha* growth, propagation, and survival to provide insight for STA operation and management. We hypothesized that (1) there is an inundation duration threshold for *Typha* in terms of survival, growth, and propagation, (2) the inundation duration threshold is longer at shallower water depths than at deeper depths and (3) inundation duration longer than the threshold reduces survival.

2. Materials and methods

2.1. Study site

This study was conducted in the north test cells of STA-1 West (STA-1 W, Fig. 1). These 15 impoundments are fully lined shallow rectangular wetlands, approximately 0.2 ha in size (Chimney et al., 2000). The cells were constructed with independent control of inflow, water depth and duration of inundation. High P water from the STA-1 W inflow canal (average P concentration > 100 μg L^{-1}) was pumped into a storage cell to supply inflow water for the study. The storage cell stage was maintained at several feet higher than the test cells. Water from the storage cell flowed by gravity into a 30-in. main distribution pipe and was delivered to each test cell through an 8-in. lateral pipe fitted with a gate valve to regulate inflow. The hydraulic loading rate (HLR) to all test cells was maintained as constant and uniform as possible throughout the study. Outflow from each test cell was controlled by a moveable 90-degree v-notch weir, which allowed water depth to be set independently within each cell.

2.2. Establishment of Typha populations

Prior to establishment of *Typha* populations within the test cells, each cell was refurbished by adding a new layer of organic soil collected from an adjacent area within STA-1 W. This soil was from an old sugarcane farm with a shallow organic layer underlain by limestone. All cells were initially inundated to prevent the establishment of unwanted terrestrial plant species. Subsequently, all cells were drained and seeded with *Typha* seeds collected from another Everglades STA during the first week of May 2018. In addition, approximately 2100 *Typha* seedlings grown in a nursery were transplanted from mid-May to mid-June 2018 at a density of 140 plants per test cell to improve the uniformity of *Typha's* spatial distribution within each cell. Water levels in the test cells during this period were maintained at soil saturation up to a few centimeters above the soil surface to support *Typha* seed germination and facilitate growth of the transplanted seedlings. Finally, water level during July 2018 to June 2019 was maintained at a depth ranging from 35 to 40 cm to encourage *Typha* growth and to suppress terrestrial weed germination. *Typha* plants were mature, i.e., > 1.5 m in height, when water-level treatments began on July 1, 2019. There was volunteer recruitment of some submerged aquatic vegetation (SAV) taxa, including *Chara* spp., *Najas guadalupensis,* and *Hydrilla verticillata*, and floating plants including *Salvinia minima*, *Azolla caroliniana* and *Lemna* spp. in all test cells, most likely via propagules transported with the inflow water.

2.3. Experimental design and inundation treatments

The study consisted of five water-level treatments: water depths of 40, 61, 84, 104, and 124 cm above the soil surface, each replicated three times, that were assigned randomly to the 15 test cells. The control treatment (40 cm) is the approximate target depth maintained in the STAs except during high inflows when treating runoff from large storms (Goforth, 2005). The 61- and 84-cm depth treatments represented shallow and moderate flooding and the 104- and 124-cm depth treatments represented deep and extremely-deep flooding, respectively. This depth range is commonly observed in the STAs and 124 cm is the maximum depth attainable in the test cells. Water depth was maintained in each treatment for 10 months from July 2019 to April 2020 and data were collected at bi-weekly or monthly intervals to evaluate changes in the *Typha* populations. *Typha* adults and juveniles were compared and analyzed in the study because juveniles were more significantly affected than adults at all water level treatments.

2.4. Typha population response

To examine the effect of water-level treatments on *Typha*, five 1 m × 1 m study plots were randomly established within each test cell using a stratified-random design that excluded areas with few or no *Typha* plants compared to the rest of the cell (Fig. 1). Plot corners were marked with PVC poles. Two baseline samplings were conducted in May and June 2019, prior to the initiation of water-level treatments on July 1, 2019, to document initial conditions of the *Typha* populations. *Typha* surveys were conducted bi-weekly for the first three months of the study and monthly thereafter, which were sufficient to characterize major phenological changes. Shoot density of adult and juvenile *Typha* ramets, and leaf elongation rate (LER) for adult ramets were monitored. Observations of visible leaf damage, presence of floating (i.e., uprooted) *Typha* in the plots, presence of other emergent or floating aquatic plant species were noted during each survey. Photos were taken of each plot during all surveys.

Typha shoot density was defined as the number of shoots within each plot (shoots m^{-2}), where ramets were categorized as 1) live adults ($>$ 1.5 m in height; with fully developed aboveground foliage), and 2) live juveniles ($<$ 1.5 m in height; with thin foliage not fully developed). In addition, five newly emerged leaves from five different adult *Typha* ramets were labeled with flagging tape marked with an identification number in each plot to monitor LER, during each monitoring period. Leaf height from the base of the ramets to the tip of each leaf was measured with a PVC pole graduated in cm. The first measurement was taken on the day the leaves were tagged, and a second measurement at 7 to 11 days thereafter. LER was calculated by dividing the change in leaf height by the number of days between measurements and expressed as cm day^{-1}.

2.5. Typha leaf photosynthesis and stomatal conductance measurements

Leaf gas exchange rates were measured using a LI-6400XT Portable Photosynthesis System (Li-COR, Lincoln Nebraska). One mature, healthy leaf was selected for photosynthesis measurements from three of the five *Typha* ramets used in the LER measurements. Once selected for measurements, the leaf was gently manipulated so that the LI-6400XT chamber could be attached 15 to 20 cm from the tip of the leaf. The chamber was held in place to allow the gas exchange measurement to stabilize before collecting the reading. This process was repeated for each selected *Typha* leaf.

2.6. Typha biomass sampling

At the end of the study, a 0.16 m^2 (0.4 m × 0.4 m) quadrat constructed from PVC pipe was positioned inside the 1 m^2 plots to collect a representative sample of *Typha* biomass including live aboveground and

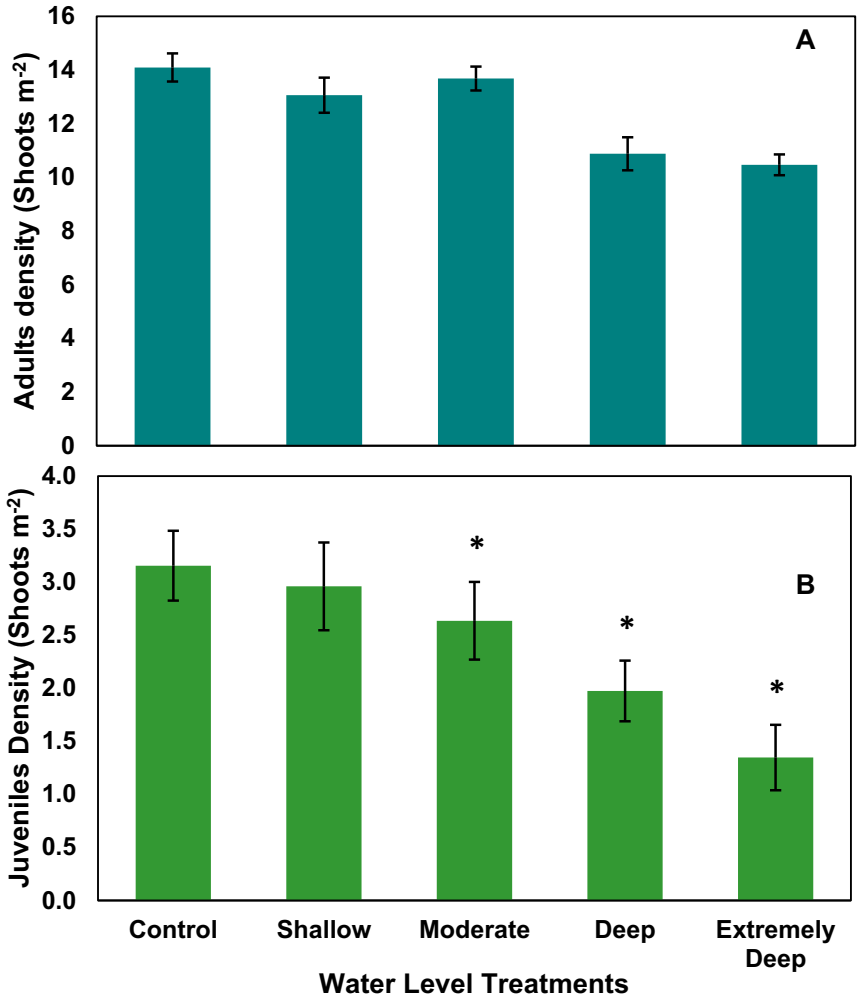


Fig. 2. Mean A) adult and B) juvenile *Typha* densities in water-level treatments in the STA-1 W test cells from July 2019 to April 2020. Asterisks indicate treatments that are significantly different at the 0.05 probability level from the control based on a one-way repeated measures ANOVA followed by a *post-hoc* mean separation test (Dunnet's). Spreader bars represent ±1 standard error from the mean.

belowground material. The sampling was done in order of water-level treatments starting with the control cells. Water depth was lowered below 20 cm the day before the scheduled sampling so that all the biomass in plots from that treatment's three test cells could be easily collected in the shortest period. Belowground biomass was removed using a long-bladed root pruning shovel to cut through the soil along the perimeter of the quadrat to a depth of approximately 30 cm. All the soil and plant material within the quadrat was carefully removed to keep ramets attached to their rhizomes. All plant-recognizable biomass was separated from this soil layer, washed on site with water from the test cell, placed in pre-labeled burlap bags for transport to the laboratory, and stored at 4 °C until biomass processing.

Typha biomass samples were processed within seven days of harvesting. Biomass materials were carefully rewashed to remove any remaining soil using a garden hose with a high-pressure nozzle. Live *Typha* ramets from each quadrat were classified as adults or juveniles and the length of the longest leaf from each ramet measured for plant height. Washed biomass from each quadrat was separated into four components: live leaves, live roots, live rhizomes, and live shoot bases. To evaluate biomass allocation among plant parts with different functions, live components were separated by ramet, cut into small pieces, placed into paper bags, and dried at 70 °C for a minimum of two weeks until a constant dry weight was obtained. Total *Typha* biomass was defined as the combined dry weight of all live tissues from live ramets. The biomass of all plant components was standardized to a unit area measurement (g m^{-2}). Biomass allocation was calculated as the dry-weight biomass of each live plant component divided by the total biomass of *Typha* in the quadrat expressed as a percent.

2.7. Data analysis

Changes in *Typha* shoot density and LER were analyzed with R and functions in the base R packages (version 4.03; R Core Team, 2020) unless otherwise noted. Data measurements were first screened for outliers following Tukey's Method (Tukey, 1977) and tested for normality with the *rstatix* R package (Kassambara, 2020). Field notes were consulted for each outlier to determine if there was a valid reason to remove the value. Test cells were the experimental units in this study while plots were subsamples within each test cell. Prior to analysis, measurements were averaged over all plots within each test cell by sampling event to avoid pseudoreplication of observations (Hurlbert, 1984). Water-level treatment effects on plant responses were evaluated using one-way repeated-measures analysis of variance (ANOVA) followed by *post-hoc* mean separation tests (Dunnett's) when the ANOVAs were statistically significant using the *lme4* and *lmerTest* R packages (Bates et al., 2015; Kuznetsova et al., 2017). Tests of model assumptions including normality and outlier assessment of standardized data were satisfactory in all cases. Mean separation tests were used to compare plant responses to water-level treatments with the control. Spearman's rank correlation coefficients (ρ) were calculated between the control and

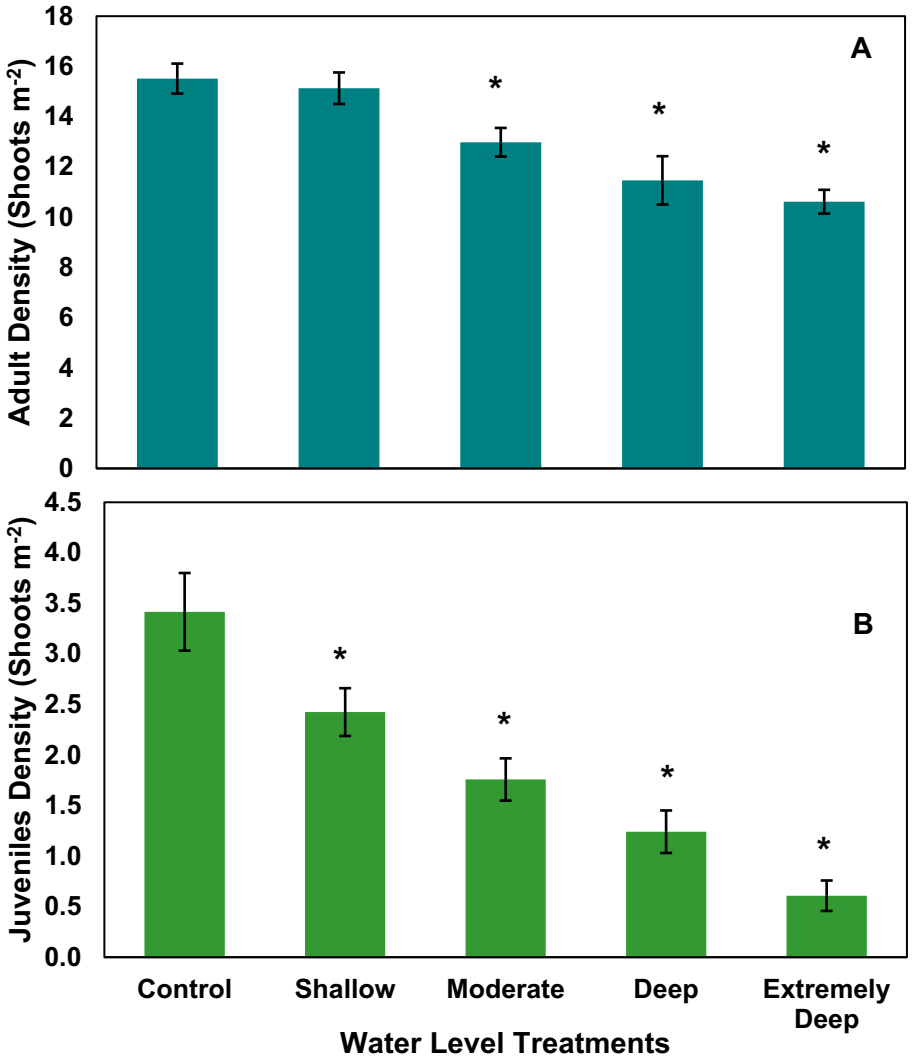


Fig. 3. Mean A) adult and B) juvenile *Typha* densities in water-level treatments in the STA-1 W test cells from July to October 2019 (i.e., the first period). Asterisks indicate treatments that are significantly different at the 0.05 probability level from the control based on a one-way repeated measures ANOVA followed by a *post-hoc* mean separation test (Dunnet's). Spreader bars represent ±1 standard error from the mean.

water-level treatment data from each biweekly/monthly monitoring event to determine if water-level treatment trends varied from the control over time. Tests were considered statistically significant at $p \leq 0.05$.

Statistical analyses of biomass data collected at the end of the study were performed using JMP® statistical software (Version 15.1.0; SAS Institute Inc. Cary, North Carolina). Data were first tested for normality (Shapiro-Wilk test) and equal variances. Treatment means of parameters that met these assumptions were compared using one-way ANOVAs to test for significant differences across water-level treatments. Parameters that did not meet these assumptions were compared using the Kruskal-Wallis Wilcoxon Rank Sum test. For mean separation comparisons, the parameters that met the assumptions for parametric analysis were analyzed using the Tukey-Kramer Honest Significant Difference test. Parameters that did not meet the assumptions were analyzed using the non-parametric multiple-comparison Dunn All Pair for Joint Ranks test. Tests were considered statistically significant at $p \leq 0.05$.

This paper presents results only for analyses of the live *Typha* biomass components.

3. Results

3.1. Effect of inundation on Typha density

Adult *Typha* densities during the baseline sampling in May and June 2019 averaged 15.1 and 12.9 adults m^{-2}, respectively. Adult densities by the end of the study in April 2020 decreased from 14.1 adults m^{-2} in the control to 10.9 and 10.5 adults m^{-2} in the deep and extremely-deep treatments (Fig. 2A), respectively, which represented a 23% and 26% decline; however, these differences were not statistically significant. Changes in adult densities were more notable during the first four months of the study (i.e., first period, July–October 2019) and were analyzed separately. Adult densities from this first period were significantly lower in the moderate (13.0 adults m^{-2}), deep (11.5 adults m^{-2}), and extremely-deep (10.6 adults m^{-2}) treatments compared to the control (15.5 adults m^{-2}, Fig. 3A), which represented a 26% and 32% decline, in the two deepest treatments, respectively. Signs of plant stress were observed as early as four weeks into the study, with *Typha* leaves from deeper treatments showing a pale green color and the first two floating *Typha* observed in one of the deep treatment plots. At 12 weeks, *Typha* from the deep and extremely-deep treatments showed considerable chlorosis and a significant decline in density compared to the shallower water-level treatments. Most of the floating *Typha* were

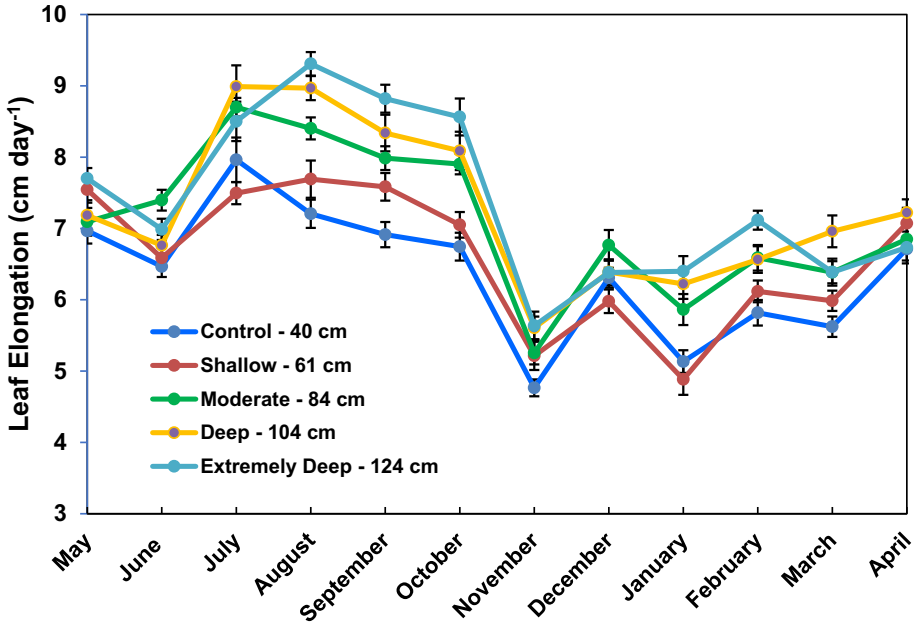


Fig. 4. Monthly mean Typha leaf elongation rates in each water-level treatment during the STA-1 W test cell study from May 2019 to April 2020. Spreader bars represent ±1 standard error from the mean.

Table 1
Spearman's rank correlation coefficients for long term trends in monthly adult and juvenile *Typha* densities, leaf elongation rate, photosynthetic rate and stomatal conductance for each water-level treatment (shallow to extremely-deep) with the control in the STA-1 W test cells. Spearman's rho (ρ) and associated *p* values are reported for each comparison. *p*-values <0.05 are statistically significant.

Typha Response	Correlation of Control with water level treatments	Spearman's ρ	Probability
Adult Density	Shallow	0.61	**0.034**
	Moderate	0.11	0.733
	Deep	0.42	0.177
	Extremely Deep	−0.17	0.604
Juvenile Density	Shallow	0.77	<0.01
	Moderate	0.42	0.170
	Deep	0.37	0.240
	Extremely Deep	0.35	0.270
Leaf Elongation Rate	Shallow	0.97	**<0.01**
	Moderate	0.94	**<0.01**
	Deep	0.87	**<0.01**
	Extremely Deep	0.85	**<0.01**
Photosynthetic Rate	Shallow	0.47	0.128
	Moderate	0.11	0.733
	Deep	0.29	0.366
	Extremely Deep	0.45	0.140
Stomatal Conductance	Shallow	0.01	0.991
	Moderate	0.45	0.140
	Deep	0.22	0.485
	Extremely Deep	0.53	0.079

observed after three months of continuous inundation. Floating *Typha* occurrence ranged from 0.13 floating *Typha* m^{-2} in the shallow to 0.27 and 0.86 floating *Typha* m^{-2} in the extremely-deep and deep treatments, respectively, however, only the floating *Typha* occurrence in the deep treatment was significantly higher compared to the rest of the treatments.

Juvenile *Typha* densities during the baseline sampling in June 2019 averaged 3.9 juveniles m^{-2}. Juvenile densities from July 2019 to April 2020 were affected by water depth, with densities declining significantly from 3.2 juveniles m^{-2} in the control to 2.0 and 1.4 juveniles m^{-2} in the deep and extremely-deep treatments, respectively, which represented a 37% and 57% decline in these treatments (Fig. 2B). *Typha* juveniles were more susceptible to deeper water-level treatments, with densities significantly declining in the deep (1.2 juveniles m^{-2}) and extremely-deep (0.6 juveniles m^{-2}) treatments compared to the control (3.4 juveniles m^{-2}, Fig. 3B) after the first four months of continuous inundation, which represented a 65% and 83% population decline in these two treatments, respectively.

3.2. Effect of inundation on Typha leaf elongation rate

Baseline LER measurements during May and June 2019 averaged 7.30 and 6.84 cm day^{-1}, respectively, when monthly water depth in the test cells ranged from 41 to 48 cm. *Typha* population rapidly responded to the different water-level treatments, with the highest monthly LERs measured after 8 weeks of continuous inundation (August 2019, Fig. 4). Trends in monthly LER varied throughout the study with all water-level treatments strongly correlated with the control ($\rho \geq 0.85$, $p < 0.001$; Table 1) and increased from 6.84 cm day^{-1} in the June 2019 baseline sampling to 8.60 cm day^{-1} in the deep and extremely-deep treatments after four weeks of inundation (late July 2019 sampling). LER remained comparatively stable in all treatments from July to October 2019 (Fig. 4). However, average LER in all treatments decreased considerably from 7.67 in October to 5.30 cm day^{-1} in November. The first two cold fronts of the season in South Florida occurred on November 17th and 25th when night-time temperatures decreased to 12 °C. These cool temperatures likely played a role in the sudden LER reduction at this time. LER increased to 6.37 cm day^{-1} in December before declining again in January 2020 and gradually increasing to near baseline levels by the end of the study.

LERs from July 2019 to April 2020 were affected by all the water-level treatments ($F = 10.0$, $p = 0.001$), with rates significantly increasing from 6.47 cm day^{-1} in the control to 7.42 cm day^{-1} in the

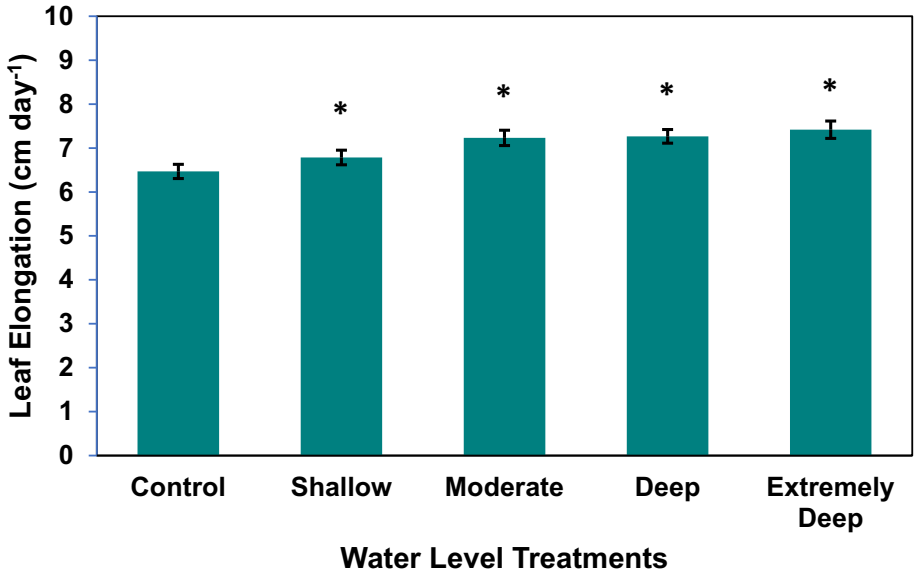


Fig. 5. Mean *Typha* leaf elongation rates in water-levels treatments from July 2019 to April 2020 in the STA-1 W test cells. Asterisks indicate that treatments are significantly different at the 0.05 probability level from the control based on a one-way repeated measures ANOVA followed by a *post-hoc* mean separation test (Dunnet's). Spreader bars represent ±1 standard error from the mean.

Table 2
Typha elongation rate (means ±1 standard error) from the first four and last six months of the STA-1 W test cell study. Asterisks indicate treatments that are significantly different at the 0.05 probability level from the control based on a one-way repeated measures ANOVA followed by a *post-hoc* mean separation test (Dunnet's).

Water Level Treatment	LER July to October 2019 (cm day^{-1})	LER November 2019 to April 2020 (cm day^{-1})
Control	7.20 ± 0.18	5.73 ± 0.17
Shallow	7.45 ± 0.12	5.98 ± 0.22
Moderate	8.25 ± 0.13*	6.35 ± 0.17*
Deep	8.59 ± 0.15*	6.47 ± 0.15*
Extremely Deep	8.77 ± 0.17*	6.42 ± 0.13*

extremely-deep treatment (Fig. 5). As with adult *Typha* density, the effect of water-level treatments on LER was more pronounced during the first four months of the study (Fig. 4) with values ranging from 7.20 cm day^{-1} in the control to 8.77 cm day^{-1} in the extremely-deep treatment (Table 2). The LERs from the moderate to extremely-deep treatments were significantly greater ($p < 0.05$) compared to the control. Monthly LERs decreased considerably during November 2019 to April 2020 (Fig. 4), with mean values ranging from 5.73 cm day^{-1} in the control to 6.47 cm day^{-1} in the deep treatment. LERs from the moderate, deep and extremely-deep treatments were significantly higher compared to the control (Table 2).

3.3. Effect of inundation on Typha gas exchange rates

Trends in photosynthetic rate and stomatal conductance (transpiration) were highly variable among the different water-level treatments but similar in terms of the timing of peak rates (Fig. 6). Photosynthetic rates and stomatal conductance declined between June and August before peaking in October with the highest values for both parameters in the deep and extremely deep-water level treatments (34.9 μmol CO_2 m^{-2} s^{-1} and 0.74 mol H_2O m^{-2} s^{-1} for photosynthetic rate and stomatal conductance, respectively; Fig. 6A and B). None of the trends in both parameters among water-level treatments were correlated with the control ($p > 0.05$; Table 1).

Photosynthetic rates varied by <3 μmol CO_2 m^{-2} s^{-1} across water-level treatments, ranging from 26.2 and 24.0 μmol CO_2 m^{-2} s^{-1} in the deep and extremely deep treatments, respectively. However, water-depth effects were significant for photosynthetic rate ($p = 0.01$), which was attributed to the significant difference between the control and the extremely deep-water level treatment (Fig. 7A). Differences in stomatal conductance rates across water-level treatments during the entire study were not statistically significant (Fig. 7B). Stomatal conductance varied by only 0.054 mol H_2O m^{-2} s^{-1} from the control (0.568 mol H_2O m^{-2} s^{-1}) to the moderate water level treatment (0.514 mol H_2O m^{-2} s^{-1}).

3.4. Effects of inundation on Typha at the end of the study

3.4.1. Typha biomass

Biomass of leaves, roots, rhizomes, and shoot bases varied among water-level treatments, but none of the among-treatment differences for each component were statistically significant at the end of the study (Table 3). Belowground biomass ranged from 1498 g m^{-2} in the control to 1295 g m^{-2} in the deep treatment. Shoot base was the only belowground component with higher biomass in the control plots compared to the treatments, which ranged from 787 g m^{-2} in the control to 603 g m^{-2} in the deep treatment. Belowground biomass: leaf ratio decreased from 0.61 in the control to 0.59 in the deeper treatments; however, the differences were not statistically significant.

Typha ramet height and biomass weight increased with increasing water depth at the end of the study, i.e., fewer, but significantly larger ramets, were found in the deeper treatments ($p < 0.05$, Fig. 8A). Adult ramet height averaged 343 and 374 cm in the deep and extremely-deep treatments, respectively, compared to 285 and 312 cm in the control and shallow treatment, respectively. Likewise, the average ramet biomass was significantly greater in the deeper treatments where the average weight of ramets from the extremely-deep treatment weighed twice that of ramets in the control (256 versus 123 g $ramet^{-1}$ respectively, Fig. 8B).

3.4.2. Typha biomass allocation

Overall, biomass allocations to leaves, roots, rhizomes, and shoot bases at the end of the study were not significantly different among water-level treatments (Fig. 9). *Typha* in all treatments allocated approximately 64% of its total biomass to leaves and 36% to belowground tissues. *Typha* in the control plots had somewhat higher shoot base allocation (19%) and lower leaf allocation (63%) than ramets in the

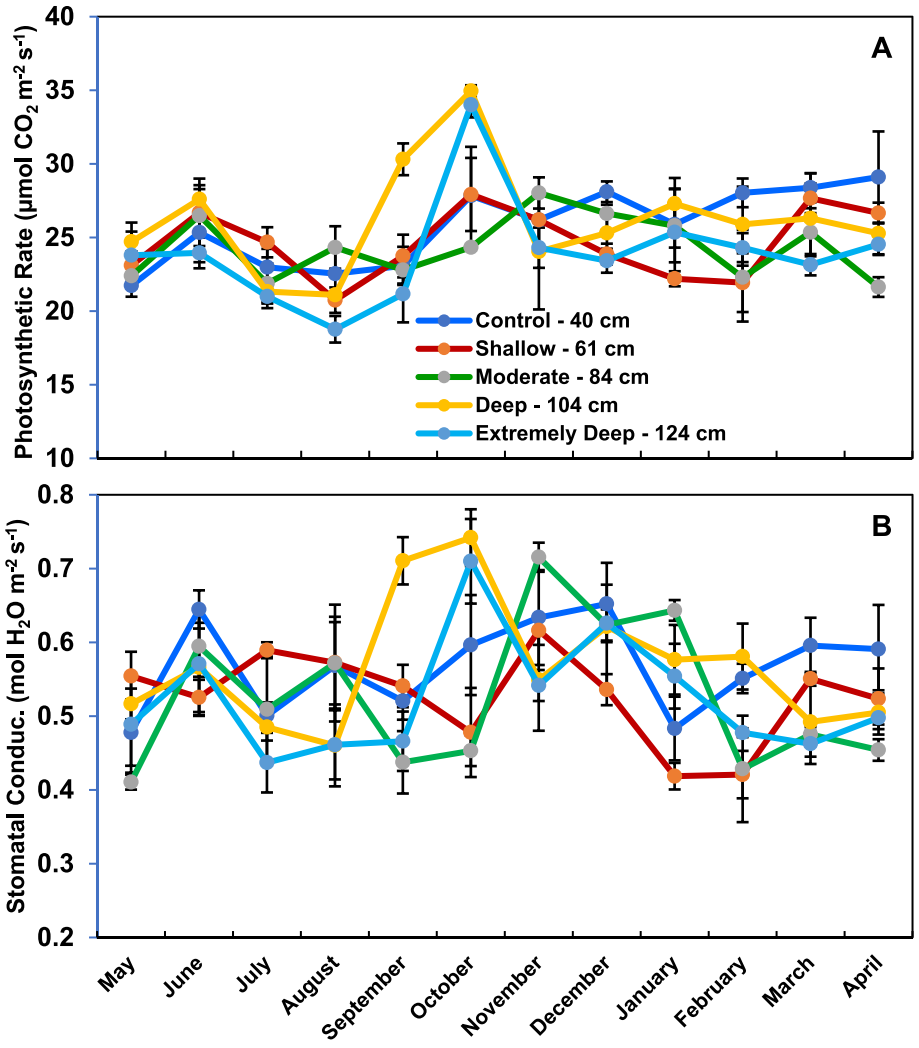


Fig. 6. Monthly means of A) photosynthetic rate and B) stomatal conductance in each water-level treatment during the STA-1 W test cells from May 2019 to April 2020. Spreader bars represent ±1 standard error from the mean.

moderate treatment (14 and 67%, respectively), whereas there was little difference among ramets from the deeper treatments.

3.4.3. *Typha* lodging

At the end of the 10 month experiment, after water was drained from the test cells, *Typha* in the moderate, deep and extremely-deep water treatments suffered from lodging, i.e. collapse of the entire plant because elongated leaves were too weak to hold the plant upright without overlying water buoyancy (Fig. 10). Lodging did not occur in the control (Fig. 10A) or shallow water treatment. The percent of plants experiencing lodging after water was drained was 10 to 15% in the moderate treatment (Fig. 10B), 50% in the deep treatment (Fig. 10C) and 60% in the extremely-deep (Fig. 10D) treatment.

4. Discussion

This study's hypotheses focused on the effects of water depth and inundation duration on *Typha*. None of the study's water-depth treatments experienced a total *Typha* population collapse by the end of the 10-month inundation period. Since none of the test cells experienced an inundation duration threshold, i.e., the time required for a total population collapse, we cannot fully evaluate our hypotheses. However, several plant responses indicated that the *Typha* communities were stressed in the deeper treatments with implications for long-term population sustainability. These responses and their implications for STA management are discussed below.

4.1. *Effect of inundation on Typha adult and juvenile densities*

Water depth exceeding a critical threshold and maintained over long periods results in physiological stress that reduces growth, biomass, density, and anchorage capacity of *Typha* (Chen and Vaughan, 2014; Miao and Zou, 2012). High flow events in the STAs that occur during South Florida's wet season result in such stressful conditions that can last for weeks if not months. However, *Typha* has demonstrated the ability to survive water depths >91 cm for several weeks in the field (Diaz and Vaughan, 2019). Surprisingly, a total *Typha* population collapse did not occur in any of our water-level treatments after 10 months of continuous inundation, as was demonstrated previously in a mesocosm study conducted over a shorter timeframe (6 weeks) at a slightly deeper water level (137 cm) (Chen et al., 2010). Increased water depth in this study did stress the *Typha* population, which was reflected in a decline in adult and juvenile densities, juvenile recruitment, increased LER, and an increase in the occurrence of floating *Typha*, mainly in the deeper treatments.

Adult *Typha* density decline in each water-level treatment was significantly different compared to the control in the first four months of the study, but not for the full study. Increasing water depths resulted in

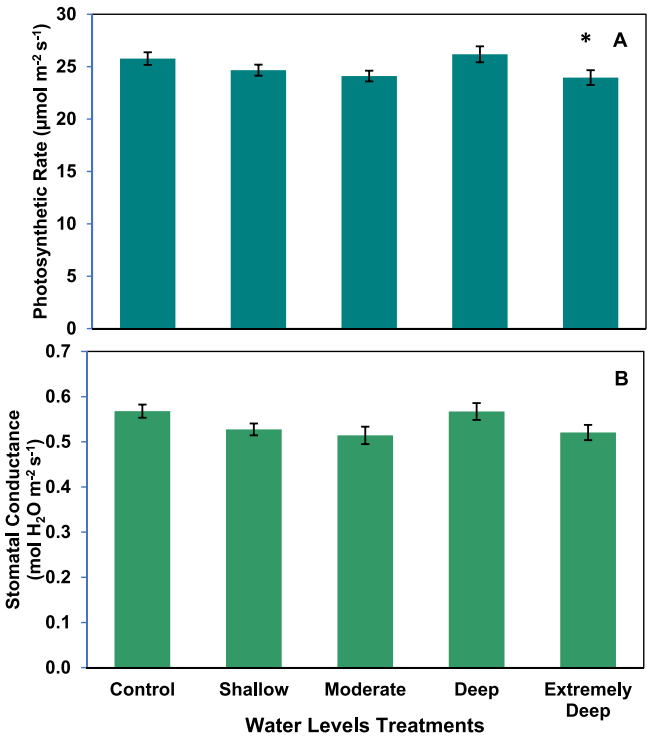


Fig. 7. Mean A) photosynthetic rate and B) stomatal conductance in water-level treatment in the STA-1 W test cells from July 2019 to April 2020. Asterisks indicate treatments that are significantly different at the 0.05 probability level from the control based on a one-way repeated measures ANOVA followed by *post-hoc* mean separation test (Dunnet's). Spreader bars represent ±1 standard error from the mean.

higher incidence of floating *Typha* as plants responded to deeper waters by decreasing biomass allocation to roots and rhizomes and increasing allocation to shoots to adapt to deeper water (Grace, 1989; Voesenek et al., 2003; Miao and Zou, 2012). Floating *Typha*, which contributed to declines in rooted-plant density, primarily occurred in the deep and extremely-deep water-level treatments after the first three months of continuous inundation.

The decline in *Typha* density was more pronounced in the juvenile populations across all water-level treatments with significant declines observed at water depths ≥84 cm (moderate depth; Figs. 2B and 3B). Although we did not observe an inundation duration threshold for *Typha* survival in any water-level treatment, water depths ≥84 cm reduced the ability of the *Typha* population to vegetatively propagate. Lower adult densities in the deeper treatments were attributed to reduced replacement of adult plants, as they died or floated over time, resulting from the decline in juvenile plant recruitment. This decreased the number of new individuals that could mature into adults. The juvenile population in this study was recruited primarily by asexual reproduction where rhizome tips transformed into new ramets (Grace and Harrison, 1986). In South Florida, new ramets can emerge year-round, especially during the spring and early summer (Weisner and Miao, 2004) fueled by energy-rich carbohydrates stored in their rhizomes (Chen et al., 2013). New ramets likely received adequate light to support growth in the control and shallow treatment (40 and 61 cm, respectively) than in the deeper treatments. These findings agree with Diaz and Vaughan (2019) who reported lower densities of adult and juvenile *Typha* in the deeper inlet region compared to the shallower outlet region of STA-3/4 Cell 2A and Grace (1989) who found that prolonged inundation adversely affected growth and reproduction of *Typha*.

4.2. Adaptation of Typha to deep inundation conditions

Typha in this study had the morphological plasticity to adapt to increased inundation to survive. Deep-water treatments stimulated *Typha* growth of younger leaves. As water depth increased, *Typha* height increased significantly, with fewer but larger, heavier ramets found in the deeper treatments. This response of *Typha* adapting to prolonged inundation by allocating more resources to the aboveground parts is consistent with results from other studies (Grace, 1989; Blom and Voesenek, 1996; Voesenek et al., 2003; Diaz and Vaughan, 2019). LER was significantly greater in *Typha* from the deeper treatments, with the highest rates observed during the most active *Typha* growing period (July–October). LER increased from 6.84 cm day^{-1} during the June baseline sampling to an average of 8.89 cm day^{-1} in the deep and extremely-deep treatments after eight weeks of continuous inundation. LER responses to the different water-level treatments may, in part, explain why *Typha* communities from the deep and extremely-deep treatments were not stressed to the point where the damage was irreversible and lethal to the plants.

4.3. Effect of inundation on Typha photosynthetic rate and stomatal conductance

The primary effects of increasing water depth on EAV are light limitation in the water column and reduced access to the atmosphere. Plants partially submerged in deep-water have less of their aboveground biomass capable of photosynthetic activity which can lead to oxygen and energy deficits due to limited aerobic metabolism (Crawford and Braendle, 1996), and reduced gas exchange between leaves and the atmosphere. *Typha* plants in this study were able to adapt to prolonged deep-water conditions through changes in morphology by growing taller via leaf elongation, facilitating the gas exchange between the above and below ground parts (Armstrong, 1979; Brix, 1993). This response of wetland plants through the development of aerenchyma or intercellular gas spaces to supply oxygen from shoots to belowground tissues (Brix et al., 1992; Bendix et al., 1994) is the most effective adaptation to survive flooded environments (Armstrong, 1979; Armstrong et al., 1994; Blom and Voesenek, 1996). Photosynthetic rates across water-level treatments during the 10 months study were significantly lower only in the extremely deep treatment compared to the control. This may partially explain the limited effect of water depth on gas exchange rates

Table 3
Mean biomass ±1 standard error of live *Typha* components for each water-level treatment at the end of the STA-1 W test cell study [a].

Water-Level Treatments	Live Components (g m^{-2})				Belowground Biomass	Aboveground Biomass	Total Live Biomass	Belowground/ Leaf Ratio
	Leaves	Roots	Rhizomes	Shoot Base	(g m^{-2})	(g m^{-2})	(g m^{-2})	
Control	2601 ± 211	363 ± 26	348 ± 24	787 ± 91	1498 ± 126	2640 ± 214	4138 ± 309	0.61 ± 0.05
Shallow	2363 ± 281	306 ± 15	346 ± 21	578 ± 40	1229 ± 64	2387 ± 283	3616 ± 334	0.65 ± 0.09
Moderate	2906 ± 291	351 ± 16	372 ± 28	615 ± 67	1338 ± 92	3047 ± 305	4385 ± 377	0.50 ± 0.04
Deep	2480 ± 338	343 ± 36	348 ± 32	603 ± 76	1295 ± 131	2543 ± 360	3838 ± 476	0.59 ± 0.06
Extremely- Deep	2863 ± 386	364 ± 22	384 ± 30	633 ± 49	1381 ± 87	2943 ± 400	4324 ± 480	0.59 ± 0.07

[a] None of the means for individual live *Typha domingensis* components were significantly different among water-level treatments at the 0.05 probability level based on the Tukey-Kramer Honestly Significant Difference test.

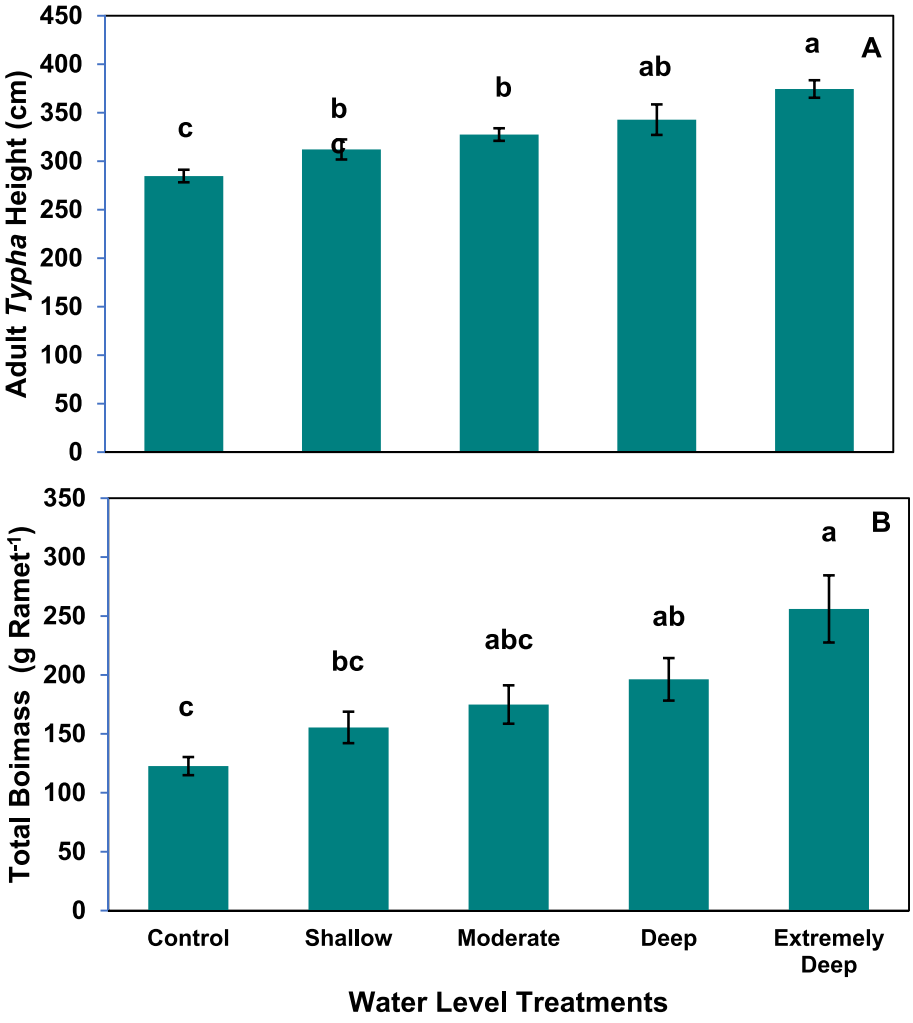


Fig. 8. Mean A) adult height and B) total biomass per *Typha* ramet as a function of water-level treatments during the final biomass harvest at the end of the STA-1 W test cell study. Treatments with the same letter are not significantly different at the 0.05 probability level based on a one-way ANOVA followed by a *post-hoc* mean separation test (Tukey-Kramer Honestly Significant Difference) for adult height and a Dunn All Pair for Joint Ranks test for total biomass. Spreader bars represent ±1 standard error from the mean.

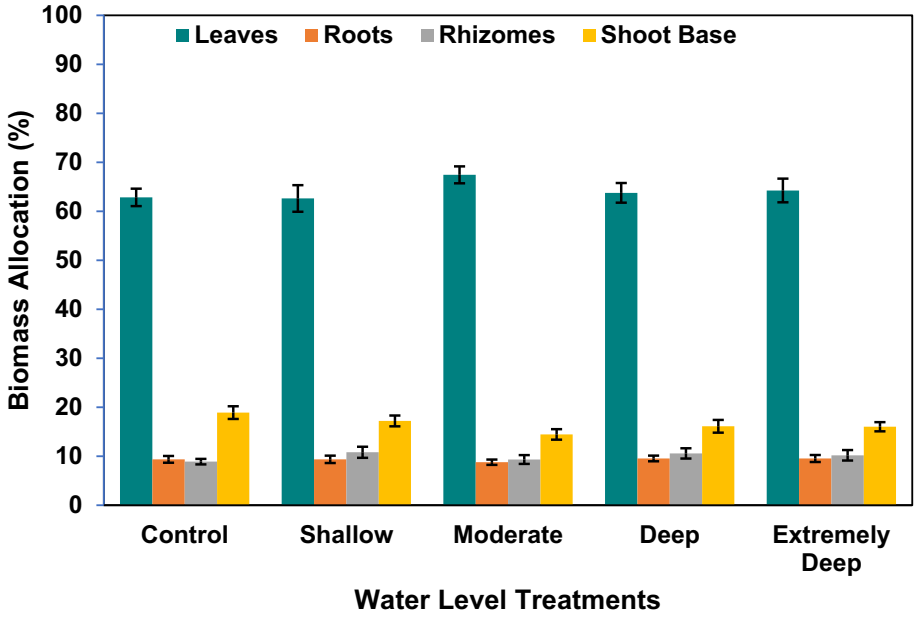


Fig. 9. Proportional biomass allocation to Typha leaves, roots, rhizomes, and shoot bases in each water-level treatment at the end of the STA-1 W test cell study. Spreader bars represent ±1 standard error from the mean.

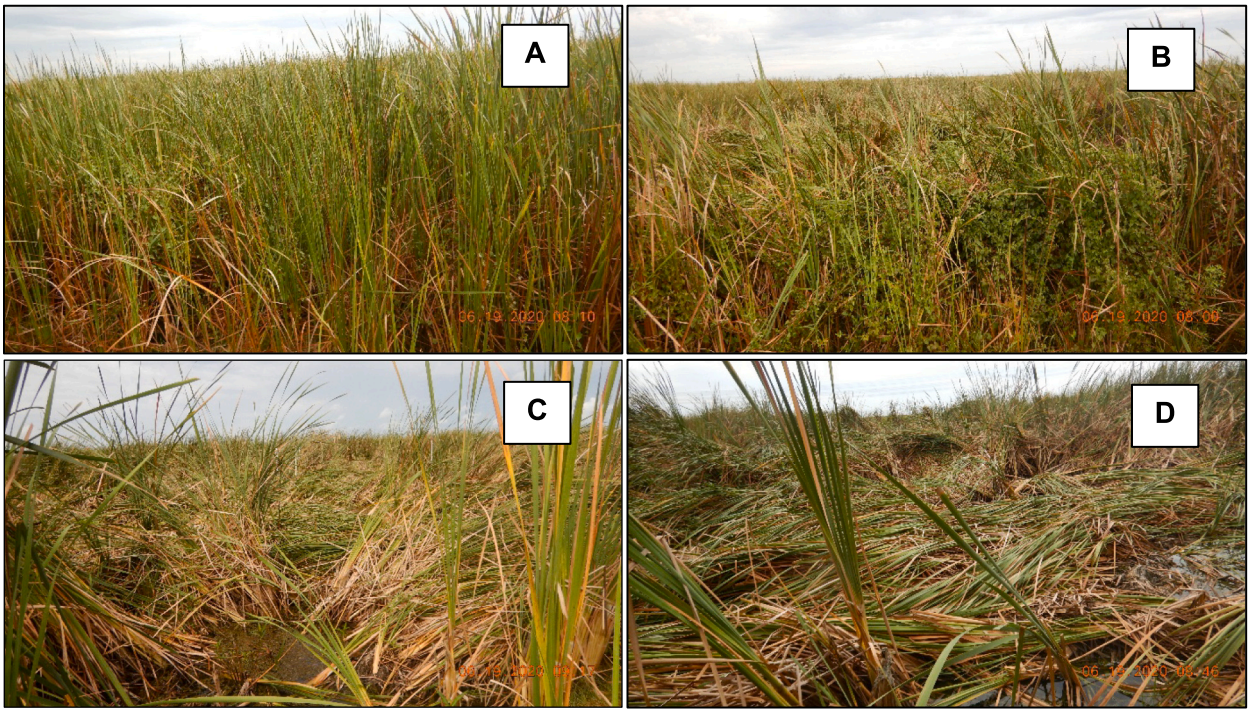


Fig. 10. Photos taken at the end of the STA-1 W test cell study showing different degrees of *Typha* lodging after the water level had been lowered below 20 cm to facilitate harvesting of all above and belowground biomass from the study plots. Photos depict A) control (40 cm), B) moderate (84 cm), C) deep (104 cm), and D) extremely-deep (124 cm) water -level treatments.

in this study. In contrast, Chen et al. (2010) and Chen and Vaughan (2014) reported significant decreases in photosynthetic rates in *T. domingensis* growing at water depths higher than 80 cm.

Stomatal conductance in EAV tends to increase toward the end of the growing season and decrease during the winter months (Mann and Wetzel, 1999). However, the small variation in stomatal conductance across water-level treatments during the 10-month study indicates that there was no physiological reduction in gas exchange despite the more difficult growing conditions in deeper water. The limited effect of water-level treatments on stomatal conductance and photosynthetic rates indicates that *Typha* was able to adapt and was not stressed to the point where these physiological parameters were significantly affected by deeper water.

4.4. Effect of inundation on Typha biomass at the end of the study

At the end of the study, fewer but significantly larger adult ramets occurred in the deeper treatments, whereas smaller and more abundant ramets (adults and juveniles) were found in the shallower treatments. This trade-off between *Typha* density and biomass of individual ramets among water-level treatments may partially explain the lack of statistical significance of the total live biomass response to water depth at the end of the study.

Typha growing under deep-water conditions typically allocates more biomass to leaves and stems than belowground tissues and has reduced vegetative reproduction (Miao and Sklar, 1998; Chen et al., 2010). However, the biomass allocation to leaves in the deeper treatments (66%) showed little difference from that in shallower depths (63%), with no statistically significant differences among water-level treatments. As water depths increased, both vegetative reproduction and juvenile density significantly decreased. Lower adult densities in the deeper treatments resulted, in part, from a decline in the number of juvenile plants that could mature into new adults. Similarities in biomass allocation among the control, shallower and deeper treatments indicates that stressed *Typha* in deeper treatments was still allocating resources to belowground tissues. Belowground biomass: leaf ratios were not significantly affected by water-level treatments, however the lower ratios from the deeper treatments indicates that roots and rhizomes were affected more than shoots. At four weeks of continuous inundation, *Typha* from the deep and extremely deep treatments started to show signs of chlorosis and the first floating *Typha* was observed in the deep treatment. Chen et al. (2010) reported that increasing water depth from 40 to 137 cm for 6 weeks in a mesocosm study significantly decreased belowground biomass: leaf ratios demonstrating that roots and rhizomes of *T. domingensis* were more affected than shoots in the deepest treatment.

Typha leaves elongated rapidly in response to changes in water depth, especially in the deep and extremely-deep treatments, where *Typha* grew taller to adapt to the deep-water conditions. During inundation induced leaf elongation, no new cells are formed, rather the increase in leaf length is a function of cell elongation (Voesenek et al., 2003), a process where the cell walls in stems and leaves become weaker allowing the cells to lengthen (Purves et al., 2004). Cell elongation reduces the structural support provided by the stems. This was most evident in the deep and extremely-deep treatments at the end of the study. During the 10-month inundation, the deeper treatments provided enough buoyancy to hold the weakened *Typha* upright. However, after water levels were lowered, this buoyancy was lost, and the stems were not able to keep most plants upright resulting in substantial *Typha* lodging. Although none of the water-level treatments experienced a total collapse of their *Typha* population by the end of the study, *Typha* lodging in the deeper treatments has important implications for managing STA water levels to keep *Typha* populations healthy.

Some of the operational challenges for STA management are maintaining optimal water levels for plant community health regardless of the magnitude and duration of inflows and establishing the range of those optimal depths. During high-flow events, vegetation density, plant community architecture, and microtopography are important parameters contributing to flow resistance (Lal, 2017; Lal et al., 2015), which in STA treatment cells with dense *Typha* stands can experience increased

water depths at their inlet regions (Diaz and Vaughan, 2019). The frequency of extreme rainfall events is expected to increase in Florida following projected climate changes (Raghavendra et al., 2018), which could increase the magnitude and duration of *Typha* inundation in the STAs. Recently constructed water storage reservoirs (i.e., Flow Equalization Basins) located upstream of the Everglades STAs are intended to reduce the adverse effects of high flows from storm events including prolonged deep-water conditions at the inlet regions of the STAs.

5. Conclusions

When *Typha* is stressed by high water levels, it is expected that plant density, biomass production and their ability to propagate will decline and the occurrence of floating *Typha* will increase. We hypothesized that increased water depth and duration of inundation would negatively affect *Typha* sustainability, i.e., continued growth and reproduction. *Typha* showed signs of stress, as evident by the decline of adult and juvenile densities and increased formation of floating *Typha* tussocks, especially in the deep and extremely-deep treatments. For the full 10 months of the study, adult densities declined by >23% in the deeper treatments; however, only juvenile densities significantly declined at water depths ≥84 cm in part because vegetative reproduction was significantly reduced at the deep and extremely deep treatments. Lower adult densities in the deeper treatments are attributed to reduced number of juveniles maturing into adults and increased number of floating *Typha*. *Typha* mitigated deep-water stress through increased LERs and biomass allocation to the aboveground parts of individual ramets. These factors led to taller *Typha* ramets in deep-water treatments that allowed them to maintain favorable gas exchange between above and belowground tissues. LER responses to deep water conditions in part explain why *Typha* was not stressed to the point where population damage was irreversible and lethal to the plants. However, increased LER in the deeper treatments resulted in less structural support in stems leading to substantial *Typha* lodging when water levels were lowered at the end of the study. While *Typha* tolerated water depths ≥84 cm, *Typha* adult and juvenile densities and the ability to vegetatively reproduce were significantly reduced in the moderate, deep and extremely-deep treatments. The first signs of chlorotic *Typha* plants, noted as early as four weeks into the study, and floating *Typha*, evident after the first three months of continuous inundation at water depths ≥84 cm, should be considered as warning signs of *Typha* stress when formulating water-level management strategies for the STAs.

CRediT authorship contribution statement

Orlando A. Diaz: Data curation, Writing – original draft, Visualization, Investigation. **Michael J. Chimney:** Writing – original draft, Writing – review & editing. **Nigel A. Temple:** Formal analysis, Writing – review & editing. **Kristin A. Vaughan:** Data curation, Writing – review & editing. **Hongjun Chen:** Formal analysis, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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